

DIGITAL FORENSICS INVESTIGATION

FOR CASE STUDY

BY

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1.0 Summary

The digital forensics investigation was conducted to determine whether the suspect, Shawn, possessed illegal content related to drug cultivation. The focus was on analyzing the forensic image of the suspect's device to uncover incriminating files, user activity, and intent.

Investigation Process

The forensic analysis was carried out using **Autopsy 4.2.1.0** and **FTK Imager 4.5.0.3** in the **Azure lab Environment**, following these key steps,

- File System Analysis – Examined directories for relevant files.
- File checking – Identified images, documents, and texts linked to illegal content.
- Metadata & Timestamp Examination – Verified file access, modification, and download history.
- Web History & Links Analysis – Investigated searches related to drugs and firearms.
- Executable & Software Analysis – Checked for tools that may support for the encryption of evidence.

All web links were **defanged** to prevent accidental access during forensic analysis.

The **end sector** was calculated using the start sector, file size, and **512 bytes per sector (found in the properties of FTK imager 4.5.0.3)**.

The formula used: **End Sector = Start Sector + (File Size / 512) - 1**.

Significance of Findings-

- ❖ Drug-related documents & images found in the Downloads folder indicate an intentional interest in marijuana cultivation.
- ❖ Deleted files suggest possible attempts to conceal evidence.
- ❖ VeraCrypt & Tor Browser were installed, commonly used for hiding illicit activities.
- ❖ Web history links confirm searches related to growing marijuana, supporting the claim of deliberate knowledge-seeking.

The evidence strongly suggests that the suspect knowingly accessed, stored, and possibly attempted to conceal content related to illegal drug activities. The presence of instructional PDFs, images, and web searches, along with encryption tools, indicates intent rather than accidental possession. These findings provide substantial digital proof to support further legal action.

2.0 Issue #1 - Content Related to the Offence Be Found in the Digital Forensic Image

2.1 Artifact #1 – 5.jpg

Name	5.jpg			Preview 
Type	File System			
MIME	image/jpeg			
Path	/img_drugs.E01/Users/shawn/Downloads/			
Status	Allocated			
Size	107,108 bytes	Physical Size	110592 Bytes	
MD5	842a442a2f7ffc1ab765fc748053be23			
SHA-256	dc36f32ae5c4c8e5c6d5efb201c6a680192ffdb6d297443d645929b96b022f8d			
Sector	34,768,216 – 34,768,424			
Accessed	2021-05-13 11:12:47 AWST			
Created	2021-05-13 11:12:47 AWST			
Modified	2021-03-25 09:25:06 AWST			
Changed	2021-03-25 09:25:06 AWST			
Analysis – This file is an image of some illegal drugs and leaves.				Significance – This file is relevant to the investigation because it could potentially serve as evidence of illegal activity. The image might be used to establish intent or association with drug related offenses

2.2 Artifact #2 – 4.jpg

Name	4.jpg			Preview PARTS OF THE CANNABIS PLANT 
Type	File System			
MIME	image/jpeg			
Path	/img_drugs.E01/Users/shawn/Downloads/			
Status	Allocated			
Size	182071 bytes	Physical Size	184,320 Bytes	
MD5	4d212ac6538e9821c712fd88767d91d2			
SHA-256	8f4cc1b219f5953b2070517b1b3017b5bd12a5e62e8cc4183f341b92101847b0			
Sector	34,767,856 - 34768211			
Modified	2021-03-25 09:20:57 AWST			
Accessed	2021-05-13 11:12:47 AWST			
Created	2021-05-13 11:12:47 AWST			
Changed	2021-03-25 09:20:57 AWST			
Analysis – This file is a detailed image depicting a cannabis plant and its specific mentioned parts. The image may serve as evidence of the suspect's involvement or interest in illegal drug activities.				

2.3 Artifact #3 – 1.jpg

Name	1.jpg			Preview 
Type	File System			
MIME	image/jpeg			
Path	/img_drugs.E01/Users/shawn/Downloads/			
Status	Allocated			
Size	90268 bytes	Physical Size	94,208 Bytes	
MD5	18c23ecd17a051be183a9e264991b686			
SHA-256	b7e7948030e6201dd129e9e476592ac3e4f49fc753b86e2fb8712131bdbca17c			
Sector	34,767,456 - 34767632			
Modified	2021-03-25 09:19:13 AWST			
Accessed	2021-05-13 11:12:47 AWST			
Created	2021-05-13 11:12:47 AWST			
Changed	2021-04-09 12:39:47 AWST			
Analysis – The file 1.jpg contains an image of an open area with a few plants planted. At first glance, the image does not appear to show any direct evidence of illegal activities. Further context may be needed to determine its relevance.				Significance - This could be linked to activities such as the trafficking of drugs in making it an important piece of evidence. This could be directly associated with locations where illegal drugs are grown or processed.

2.4 Artifact #4 – 2.jpg

Name	2.jpg			Preview 
Type	File System			
MIME	image/jpeg			
Path	/img_drugs.E01/Users/shawn/Downloads/			
Status	Allocated			
Size	78158 bytes	Physical Size	81,920 Bytes	
MD5	66bf4cd3df4bbddf1e92478ae0f755eb			
SHA-256	f74e1ee386c62596fb60cb277eef9e31403103f571275bdf9c223bb412f055fa			
Sector	34,767,640 - 34767792			
Modified	2021-03-25 09:19:25 AWST			
Accessed	2021-05-13 11:12:47 AWST			
Created	2021-05-13 11:12:47 AWST			
Changed	2021-03-25 09:19:25 AWST			
Analysis – This image shows a plant, potentially related to drug cultivation. Although the exact type of plant is not identifiable in the image, its presence on the suspect's computer may indicate interest with illegal activities such as drug.				Significance - Even though the plant type cannot be confirmed at this stage, it may still be used as evidence to establish a connection between the suspect and drug related activities.

2.5 Artifact #5 – 3.jpg

Name	3.jpg			Preview 
Type	File System			
MIME	image/jpeg			
Path	/img_drugs.E01/Users/shawn/Downloads/			
Status	Allocated			
Size	26624 bytes	Physical size	28,672 Bytes	
MD5	dee5d9a3462c9f331905267d2e3eb489			
SHA-256	7cd559138cd9cf918eef7d5249af38fb47e1897b5340af9c94eb82516643b48e			
Sector	34,767,800 - 34767852			
Modified	2021-03-25 09:19:47 AWST			
Accessed	2021-05-13 11:12:47 AWST			
Created	2021-05-13 11:12:47 AWST			
Changed	2021-03-25 09:19:47 AWST			
Analysis – This is an image of a plant and its part. Possibly related to cannabis or other plants that are commonly associated with drug production or illegal activities.				Significance - The presence of this file in the Downloads folder suggests intentional retrieval, possibly to further a criminal act like cultivation or trafficking.

2.6 Artifact #6 – cc.jpg

Name	cc.jpg			Preview 
Type	File System			
MIME	image/jpeg			
Path	/img_drugs.E01/Users/shawn/Downloads/			
Status	Allocated			
Size	9420 bytes	Physical Size	12,288 Bytes	
MD5	5d7fc49118f0ce7b93f1ed3b2266e523			
SHA-256	308afc2435a1817db0c3c509c5bb08f19693503d8744142756a7f7d60a02c087			
Sector	3476842 - 3476860			
Modified	2021-04-19 10:00:53 AWST			
Accessed	2021-05-13 11:12:47 AWST			
Created	2021-05-13 11:12:47 AWST			
Changed	2021-04-19 10:00:53 AWST			
Analysis – This image shows what appears to be white powdered substances arranged in lines. This is commonly associated with illegal drug such as cocaine. The appearance shows that the lines powdered could be before the intake.				Significance - This image may serve as direct evidence of drug-related activities in the context of the investigation, as it suggests the presence of powdered drugs

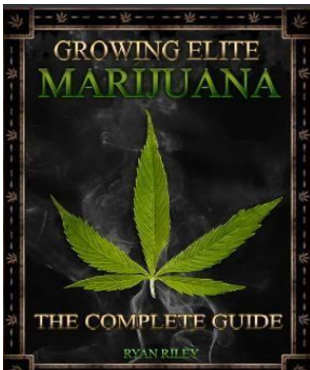
2.7 Artifact #7 – cc2.jpg

Name	cc2.jpg			Preview 
Type	File System			
MIME	image/jpeg			
Path	/img_drugs.E01/Users/shawn/Downloads/			
Status	Allocated			
Size	9119 bytes	Physical Size	12,288 Bytes	
MD5	ae6941b2f7aa5cc738b9a15005297762			
SHA-256	02b92ebec15afbb6d2126c97707275332383fbbd30361934188ca6536db215a5			
Sector	34,768,456 – 34,768,473			
Modified	2021-04-19 10:01:03 AWST			
Accessed	2021-05-13 11:12:47 AWST			Significance –This could serve as important evidence connecting the suspect to drug-related offenses, particularly in relation to the intake of drugs.
Created	2021-05-13 11:12:47 AWST			
Changed	2021-04-19 10:01:03 AWST			
Analysis - This image features rolled money, white powder lines and a card. The card is likely used to spread the white powder into lines, while the rolled money is typically used for inhaling or snorting substances. The white powder may indicate to illicit drugs such as cocaine.				

2.8 Artefact #8 – 410.jpg

Name	image410.jpg			Preview 
Type	File System			
MIME	image/jpeg			
Path	/img_drugs.E01/Users/shawn/Downloads/how-to-grow-elite-marijuana.pdf/			
Status	Allocated			
Size	61304 bytes	Physical Size	61,440 Bytes	
MD5	09584a534731621a46e3256c6946a3e2			
SHA-256	3e813937f5add960dec26f883a79297ecf4332534faf78b94ae928df262988ae			
PDF Sector	34,775,128 - 34,75246			
Modified	Timestamp Unavailable - 0000-00-00 00:00:00			Significance - It could be directly linked to the cultivation and distribution of marijuana, which is a central element in the ongoing investigation.
Accessed	Timestamp Unavailable - 0000-00-00 00:00:00			
Created	Timestamp Unavailable - 0000-00-00 00:00:00			
Changed	Timestamp Unavailable - 0000-00-00 00:00:00			
Analysis – This image contains of Hash, Pot Marijuana, and Skunk labelled. It was found in the PDF titled 'How to Grow Elite Marijuana' located in the Downloads folder.				

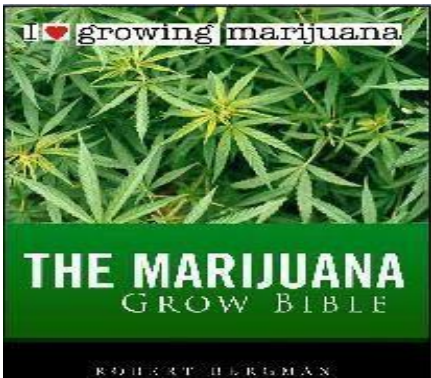
2.9 Artifact #9 – how-to-grow-elite-marijuana.pdf

Name	how-to-grow-elite-marijuana.pdf			Preview 
Type	File System			
MIME	application/pdf			
Path	/img_drugs.E01/Users/shawn/Downloads/			
Status	Allocated			
Size	38569182 bytes	Physical Size	38,572,032 Bytes	
MD5	d64aed98330277fcf2afbb1f5a4af9cc			
SHA-256	b1e96a9571d478b98c26b9f177428b40a0ac66b3b1c3d5ce1e7d899be518f3d5			
Sector	34,775,128 - 34,850458			
Modified	2021-03-25 09:40:33 AWST			
Accessed	2021-05-13 11:12:49 AWST			
Created	2021-05-13 11:12:49 AWST			
Changed	2021-04-09 11:26:28 AWST			
Analysis – This is a PDF document titled "how-to-grow-elite-marijuana.pdf," which provides detailed instructions on marijuana cultivation. The document appears to contain information about the illegal production of marijuana.				

2.10 Artifact #10 – Marijuana Grow Bible.pdf

Name	Marijuana Grow Bible.pdf		
Type	File System		
MIME	application/pdf		
Path	/img_drugs.E01/Users/shawn/Downloads/		
Status	Allocated		
Size	1,721,732 bytes	Physical Size	4,356,325 Bytes
MD5	a780f8c9b2ee5415995546ea48006fbf		
SHA-256	6c45dd7cba3059a4c0033c6aebcb9a1a025d97490663bbecbe05bbe02d0808d3		
Sector	34,850,600 - 3485396		
Modified	2021-03-25 09:40:18 AWST		
Accessed	2021-05-13 11:12:51 AWST		
Created	2021-05-13 11:12:51 AWST		
Changed	2021-04-15 14:13:48 AWST		
Analysis	– This is a PDF document titled <i>Marijuana Grow Bible.pdf</i>, likely containing instructions for the cultivation of marijuana. This document could be used as a guide for those involved in the cultivation or trafficking of illicit substances.		

Preview



Significance - The presence of such a document in the Downloads folder suggests it was intentionally downloaded, possibly related to illegal activities such as growing marijuana

3.0 Issue #2: If Illegal Content Related to the Offence Is Present, Can the Suspect Be Linked to It?

The files retrieved from the suspect's system contain various images related to illegal drug activities. These images may serve as evidence linking the suspect, Shawn, to the suspected illegal activities, including drug trafficking cultivation and consumption.

3.1 Artefacts Linked to the Offence

Artifact #1: 5.jpg

- ✓ The file is located in Shawn's folder inside the Users directory and was accessed and created on May 13, 2021, with modification on March 25, 2021. The modification timestamp suggests that Shawn may have revisited or saved this file after it was originally downloaded, indicating intentional retrieval or further involvement with the content.

Artifact #2: 4.jpg

- ✓ The image was created and accessed on May 13, 2021, and modified on March 25, 2021, which was also found in Shawn's folder inside the Users directory. The modification and access times suggest deliberate action of obtaining and interacting with this file, potentially linking user to drug-related activities.

Artifact #3: 1.jpg

- ✓ Found in Shawn's folder inside the Users directory, as this file was created and accessed on May 13, 2021, and modified on March 25, 2021. The timestamps show that user interacted with the file on the same day, possibly indicating involvement in related illegal activities.

Artifact #4: 2.jpg

- ✓ Found in Shawn's folder inside the Users directory, this file was created and accessed on May 13, 2021, and modified on March 25, 2021. The modification date suggests that user revisited or altered the file, further establishing a connection to potentially illegal activities.

Artifact #5: 3.jpg

- ✓ Like the other artefacts, this image was created and accessed on May 13, 2021, with a modification date of March 25, 2021. The file is in Shawn's folder inside the Users directory which shows continuous involvement in plants linked to illegal activities such as drug cultivation or trafficking.

All 5 files mentioned from **Artifact #1 to Artifact #5** in the above were modified on the same date **March 25, 2021**, by the same user “Shawn”, suggests that they were accessed or altered in some way on this specific day. This could indicate a batch download in which the user accessed it later by the same day.

Artifact #6: cc.jpg

- ✓ This file is located in Shawn’s folder inside the Users directory that was created and accessed on May 13, 2021, with modification on April 19, 2021.

Artifact #7: cc2.jpg

- ✓ Same as Artifact #6, Artifact #7 is found in Shawn’s folder within the Users directory. Both files were created and accessed on May 13, 2021, with modification timestamps indicating changes on April 19, 2021. While these artefacts were likely downloaded on the same day as the others mentioned earlier, the specific modification date suggests a closer connection between them.

Interestingly, both images feature **white powder**, and the **file names “cc.jpg” and “cc2.jpg”** are identical. This suggests a potential link between the two artefacts, possibly pointing to a common drug such as cocaine. The similarity in file names and content further strengthens the argument that these artefacts could be directly related to drug related activities. This consistent pattern of timestamps across different files strengthens the case that suspect was engaged in illegal activities, directly linking him to the offense.

Artifact#8 and Artifact#9

- ✓ The first image and PDF (Artifact #8 & #9) are directly related since the image is embedded in the *How to Grow Elite Marijuana.pdf*. The pdf was found in the downloads folder of the “Shawn” folder in the Users directory. This shows that use is linked as it was downloaded.

Artifact#10

- ✓ The pdf was found in the downloads folder of the “Shawn” folder in the Users directory. This shows that use is linked as it was downloaded.

4.0 Issue#3 -Was the Content Distributed Intentionally?

4.1 Artifact#9 and Artifact#10

These 2 pdf files show how to grow and maintain Marijuana. Given that the files contain specific and instructional content about marijuana cultivation, this points to the possibility that the user intentionally accessed these resources as part of their efforts to gain knowledge related to illegal activities, such as drug production or trafficking. There was no specific sector allocated for Zone Identifier files because they were embedded within the image file itself. As a result, their data was included within the image file's sectors.

4.2 Artifact#11 - 1.jpg:Zone.Identifier

Name	1.jpg:Zone.Identifier			Preview [ZoneTransfer] ZoneId=3 METADATA Significance -This shows that the 1.jpg was downloaded from an untrusted source.
Type	File System			
MIME	text/plain			
Path	/img_drugs.E01/Users/shawn/Downloads/			
Status	Allocated			
Size	26 bytes	Physical Size	32 Bytes	
MD5	fbccf14d504b7b2dbcb5a5bda75bd93b			
SHA-256	eacd09517ce90d34ba562171d15ac40d302f0e691b439f91be1b6406e25f5913			
1.jpg Sector	34,767,456 – 34.767,632			
Modified	2021-03-25 09:19:13 AWST			
Accessed	2021-05-13 11:12:47 AWST			
Created	2021-05-13 11:12:47 AWST			
Changed	2021-04-09 12:39:47 AWST			

4.3 Artifact#12 - 2.jpg:Zone.Identifier

Name	2.jpg:Zone.Identifier			Preview [ZoneTransfer] ZoneId=3 METADATA
Type	File System			
MIME	text/plain			
Path	/img_drugs.E01/Users/shawn/Downloads/			
Status	Allocated			
Size	26 bytes	Physical Size	32 Bytes	
MD5	fbccf14d504b7b2dbcb5a5bda75bd93b			
SHA-256	eacd09517ce90d34ba562171d15ac40d302f0e691b439f91be1b6406e25f5913			
2.jpg Sector	34,767,640 – 34,767,792			
Modified	2021-03-25 09:19:25 AWST			
Accessed	2021-05-13 11:12:47 AWST			
Created	2021-05-13 11:12:47 AWST			
Changed	2021-03-25 09:19:25 AWST			

Significance -This shows that the 2.jpg was downloaded from an untrusted source

4.4 Artifact#13 - 3.jpg:Zone.Identifier

Name	3.jpg:Zone.Identifier			Preview [ZoneTransfer] ZoneId=3 METADATA Significance - This shows that the 3.jpg was downloaded from an untrusted source
Type	File System			
MIME	text/plain			
Path	/img_drugs.E01/Users/shawn/Download			
Status	Allocated			
Size	26 bytes	Physical Size	32 Bytes	
MD5	fbccf14d504b7b2dbcb5a5bda75bd93b			
SHA-256	eacd09517ce90d34ba562171d15ac40d302f0e691b439f91be1b6406e25f5913			
3.jpg Sector	34,767,800 – 34,767,852			
Modified	2021-03-25 09:19:13 AWST			
Accessed	2021-05-13 11:12:47 AWST			
Created	2021-05-13 11:12:47 AWST			
Changed	2021-04-09 12:39:47 AWST			

4.5 Artifact#14 - 4.jpg:Zone.Identifier

Name	4.jpg:Zone.Identifier			Preview [ZoneTransfer] ZoneId=3 METADATA
Type	File System			
MIME	text/plain			
Path	/img_drugs.E01/Users/shawn/Downloads/			
Status	Allocated			
Size	26 bytes	Physical Size	32 Bytes	
MD5	fbccf14d504b7b2dbcb5a5bda75bd93b			
SHA-256	eacd09517ce90d34ba562171d15ac40d302f0e691b439f91be1b6406e25f5913			
4.jpg Sector	34,767,856 – 34,768,211			
Modified	2021-04-19 10:00:53 AWST			
Accessed	2021-05-13 11:12:47 AWST			
Created	2021-05-13 11:12:47 AWST			
Changed	2021-04-19 10:00:53 AWST			

Significance - This shows that the 4.jpg was downloaded from an untrusted source

4.6 Artifact#15 - 5.jpg:Zone.Identifier

Name	5.jpg:Zone.Identifier			Preview [ZoneTransfer] ZoneId=3
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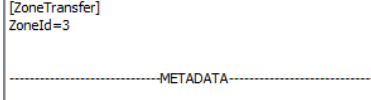
4.7 Artifact#16 - cc.jpg:Zone.Identifier

Name	cc.jpg:Zone.Identifier			Preview [ZoneTransfer] ZoneId=3 METADATA
Type	File System			
MIME	text/plain			
Path	/img_drugs.E01/Users/shawn/Downloads/			
Status	Allocated			
Size	26 bytes	Physical Size	32 Bytes	
MD5	fbccf14d504b7b2dbcb5a5bda75bd93b			
SHA-256	eacd09517ce90d34ba562171d15ac40d302f0e691b439f91be1b6406e25f5913			
cc.jpg Sector	3476842 - 3476860			
Modified	2021-04-19 10:00:53 AWST			
Accessed	2021-05-13 11:12:47 AWST			
Created	2021-05-13 11:12:47 AWST			
Changed	2021-04-19 10:00:53 AWST			

Significance

- This shows that the cc.jpg was downloaded from an untrusted source

4.8 Artifact#17 – cc2.jpg:Zone.Identifier

Name	cc2.jpg:Zone.Identifier	Preview  The preview shows the raw data of the Zone.Identifier file. It contains a header "[ZoneTransfer]" followed by "ZoneId=3". Below this, there is a dashed line labeled "METADATA".
Type	File System	
MIME	text/plain	
Path	/img_drugs.E01/Users/shawn/Downloads/	
Status	Allocated	
Size	26 bytes	
MD5	fbccf14d504b7b2dbcb5a5bda75bd93b	
SHA-256	eacd09517ce90d34ba562171d15ac40d302f0e691b439f91be1b6406e25f5913	
cc2 Sector	34,768,456 – 34,768,473	
Modified	2021-04-19 10:01:03 AWST	
Accessed	2021-05-13 11:12:47 AWST	
Created	2021-05-13 11:12:47 AWST	
Changed	2021-04-19 10:01:03 AWST	

Significance - This shows that the cc2.jpg was downloaded from an untrusted source

4.9 Artifact#11 - Artifact#17 Analysis: Zone.Identifier Files Overall Analysis

- **These Zone.Identifier files (Artifact#11 to Artifact#17)** show that the associated images were downloaded from an untrusted external source, as shown by the "[ZoneTransfer] ZoneId=3" data. This Shows the user was intentionally accessing illegal content from an untrusted source. The zone identifier files have the same MD5 Hash- "**fbccf14d504b7b2dbcb5a5bda75bd93b**" and SH1 hash – "**eacd09517ce90d34ba562171d15ac40d302f0e691b439f91be1b6406e25f5913**", showing this was a batch download from one web site.

5.0 Transaction Related Web Artifacts

Artifact#18 – Transaction Web Artifact 1

Created	2021-05-13 10:58:43 AWST	Modified	2020-12-09 12:28:47 AWST
Accessed	2021-05-13 10:58:43 AWST	Changed	2021-03-25 10:18:56 AWST
Location	/img_drugs.E01/Windows/WinSxS/amd64_microsoft-windows-crypt32-dll_31bf3856ad364e35_6.3.9600.19913_none_4c78981cb1932c2d/		
File Name	crypt32.dll		
MIME Type	application/x-msdownload		
MD5	0462b10e1bcf14fdb7b09d7b832ebd21		
SHA -256	a5c1926aaa0e74577ffd09af3d6b958e82a1dc1d912bf1c68464175b7451295b		
Keywords Found	20181209000224, \$gc31, xf#[		
Analysis	The credit card number and cryptographic reference (crypt32.dll) suggest potential illicit activity, possibly involving encrypted files or hidden data. The cryptocurrency-related keyword (\$gc31) further indicates illegal transaction and there is also credit card number “20181209000224”		

6.0 Email Titles which could be Related to Illegal Activities

Artifact#19 – Email Web Artifact 1

Domain	google.com	MIME Type	application/x-sqlite3
Date & Time	2021-03-25 09:41:38 AWST (Date Accessed)		
Location	/img_drugs.E01/Users/shawn/AppData/Local/Google/Chrome/User Data/Default/History		
MD5	6349ad7c74ea3c2268b5f30210c0f5cf		
SHA -256	28c6b44b32ea583cd8c6753cb2ce8a37e8c238e2906e85b54be7ba9b3587714e		
Title Text	Shawn, finish setting up your new Google Account - skemp19951@gmail.com - Gmail		
Program	Google Chrome		
Link	hxxps://mail.google.com/mail/u/0/#inbox/FMfcgxwLswMrnnDJJXrhNzWtcMTLSchM		
Analysis	This email “Shawn, finish setting up your new Google Account - skemp19951@gmail.com – Gmail”. subject shows that the user may have been attempting to establish an anonymous account.		

Artifact#20 – Email Web Artifact 2

Domain	google.com	MIME Type	application/x-sqlite3
Date & Time	2021-04-09 09:55:41 AWST (Date Accessed)		
MD5	6349ad7c74ea3c2268b5f30210c0f5cf		
Program	Google Chrome		
SHA -256	28c6b44b32ea583cd8c6753cb2ce8a37e8c238e2906e85b54be7ba9b3587714e		
Location	/img_drugs.E01/Users/shawn/AppData/Local/Google/Chrome/User Data/Default/History		
Title Text	Plus500 CFD: Online Trading - skemp19951@gmail.com - Gmail		
Link	https://mail.google.com/mail/u/0/#ad/298		
Analysis	The title "Plus500 CFD: Online Trading." suggests the user has been involved in using online trading platforms. By this activity and the creation of the creation of email account raises concerns about possible financial transactions related to the drug trade.		

7.0 Drugs and Transaction Web visits

All the Search Terms below were:

- Done by the Program – **Google Chrome**
- Taken from Web artifact location –
(/img_drugs.E01/Users/shawn/AppData/Local/Google/Chrome/User
Data/Default/History)

Artifact#21 – Drug Search History 1			
Domain	google.com	Date & Time	2021-03-22 09:16:46 AWST
Term	Drugs		
Analysis	The user searched for “Drugs” probably related to starting to learn on drugs.		
Artifact#22 – Drug Search Term 2			
Domain	google.com	Date & Time	2021-03-22 09:18:21 AWST
Term	popular drugs among youth		
Analysis	Searched the most famous drugs among young generation, could show that the user is investigating which drugs are in demand.		
Artifact#23 – Drug Search Term 3			
Domain	google.com	Date & Time	2021-03-22 09:19:23 AWST
Term	most common drugs among youth		
Analysis	Searched the most common drug, this could be relevant for illegal selling.		
Artifact#24 – Drug Search Term 4			
Domain	google.com	Date & Time	2021-03-22 09:20:21 AWST
Term	most common drugs among youth in australia		
Analysis	Searched the most common drug in Australia, this could be for selling illegally within the country.		
Artifact#25 – Drug Search Term 5			
Domain	google.com	Date & Time	2021-03-22 09:26:12 AWST
Term	most expensive drugs		
Analysis	Searched the most expensive drug, this could be relevant for illegal selling.		
Artifact#26 – Drug Search Term 5			
Domain	google.com	Date & Time	2021-03-22 09:27:26 AWST
Term	Marijuana		
Analysis	Searched marijuana, which could be relevant for cultivation as found in Artifacts #9 and #10.		
Artifact#27 – Drug Search Term 6			
Domain	google.com	Date & Time	2021-03-25 09:17:44 AWST
Term	different types of drugs and their prices		
Analysis	The user’s search is relevant to search about the prices to sell.		
Artifact#28 – Drug Search Term 7			
Domain	youtube.com	Date & Time	2021-04-08 09:57:02 AWST
Term	how to grow lsd, mdma and lsdhow to grow lsd, mdma and lsd		
Analysis	User searching "how to grow LSD, MDMA, and LSD" on YouTube suggests that the user’s interest in drug production.		

8.0 Drugs and Transaction Search History Web visits

All the web visits below were:

- Done by the Program – **Google Chrome**
- Taken from Web artifact location –
(/img_drugs.E01/Users/shawn/AppData/Local/Google/Chrome/User
Data/Default/History)

Artifact#29 – Drug Search History 1			
Domain	netflix.com	Date & Time	2021-04-12 14:10:33 AWST (Date Accessed)
Title	How to Sell Drugs Online (Fast) Netflix Official Site		
Link	hxxps://www.netflix[.]com/au/title/80218448		
Analysis	User has searched related content on Netflix to watch videos to learn and sell drugs online.		
Artifact#30 – Drug Search History 2			
Domain	weedmaps.com	Date & Time	2021-04-09 12:39:32 AWST (Date Accessed)
Title	The Different Parts Of A Marijuana & Cannabis Plant		
Link	hxxps://weedmaps[.]com/learn/the-plant/parts-of-cannabis-plant		
Analysis	User has visited an untrusted site to learn about drugs.		
Artifact#31 – Drug Search History 3			
Domain	thevespiary.org	Date & Time	2021-04-09 11:39:12 AWST (Date Accessed)
Title	The Cannabis Grow Bible: The definitive guide to growing marijuana for recreational and medical use.		
Link	hxxps://www[.]thevespiary.org/library/Files_Uploaded_by_Users/llamabox/Newcannabisgrowbible.pdf		
Analysis	User has visited an untrusted site and read a pdf document to learn how to grow illegal drugs.		
Artifact#32 – Drug Search History 4			
Domain	marijuanaplantsonline.com	Date & Time	2021-03-25 09:39:33 AWST (Date Accessed)
Title	Not Available		
Link	http://www.marijuanaplantsonline[.]com/downloads/how-to-grow-elite-marijuana.pdf		
Analysis	User has visited an illegal site and read a pdf document in Artifact #9 to learn about Marijuana drugs.		
Artifact#33 – Drug Search History 4			
Domain	ilovegrowingmarijuana.com	Date & Time	2021-03-25 09:38:25 AWST (Date Accessed)
Title	200+ Marijuana Grow Guides – Indoor & Outdoor Strains ILGM		
Link	hxxps://www.ilovegrowingmarijuana[.]com/guides/		
Analysis	User has visited an illegal site and read a pdf document in Artifact #9 to learn about Marijuana drugs.		
Artifact#34 – Drug Search History 6			
Domain	pharmout.net	Date & Time	2021-03-25 09:22:22 AWST (Date Accessed)
Title	The Price of Cannabis		
Link	hxxps://www.pharmout[.]net/price-of-cannabis/		
Analysis	User visited an illegal site to learn about the price of cannabis.		

9.0 Software Applications Related Search History related to help the Incident

All the web visits below were:

- Done by the Program – **Google Chrome**
- Taken from Web artifact location –
(/img_drugs.E01/Users/shawn/AppData/Local/Google/Chrome/User Data/Default/History)

Artifact#35 – Software Search History 1			
Domain	csoonline.com	Date & Time	2021-03-24 07:46:33 AWST (Date Accessed)
Title	What is the Tor Browser? And how the dark web browser works CSO Online		
Link	hxxps://www.csoonline[.]com/article/3287653/what-is-the-tor-browser-how-it-works-and-how-it-can-help-you-protect-your-identity-online.html		
Analysis	The search for Tor Browser shows the user’s intent to access dark web, for illicit activities.		
Artifact#36 – Software Search History 2			
Domain	thunderbird.net	Date & Time	2021-04-29 10:29:55 AWST (Date Accessed)
Title	Thunderbird — Make Email Easier. — Thunderbird		
Link	hxxps://www[.]thunderbird.net/en-GB/		
Analysis	The search for Thunderbird shows users’ interest in a secure email client, which may be used for encrypted communication.		
Artifact#37 – Software Search History 3			
Domain	truecrypt.org	Date & Time	2021-04-15 09:28:37 AWST (Date Accessed)
Title	TrueCrypt		
Link	hxxp://www[.]truecrypt.org/downloads		
Analysis	The search for TrueCrypt shows interest in disk encryption software, used to protect sensitive data.		
Artifact#38 – Software Search History 4			
Domain	ritlabs.com	Date & Time	2021-04-29 11:01:32 AWST (Date Accessed)
Title	Ritlabs, SRL - secure email client software for Windows		
Link	hxxp://www[.]ritlabs.com/en/products/		
Analysis	Searching another secure email client, shows a continued effort for untraceable communication		
Artifact#39 – Software Search History 4			
Domain	softonic.com	Date & Time	2021-04-09 11:56:23 AWST (Date Accessed)
Title	Download BitLocker Password - latest version		
Link	hxxps://bitlockerspassword.en.softonic[.]com/download		
Analysis	The search for BitLocker shows an effort to decrypt secure data, likely related to illegal or sensitive files.		
Artifact#40 – Software Search History 6			
Domain	okta.com	Date & Time	2021-04-30 15:27:12 AWST (Date Accessed)
Title	okta-dev-58707408 - Getting Started		
Link	hxxps://dev-58707408-admin.okta[.]com/admin/getting-started#		
Analysis	The Okta search indicates an interest in secure access tools, suggesting the user may be trying to access systems protecting sensitive data for illicit activities.		

10.0 Transaction Related Search History Web visits

All the web visits below were:

- Done by the Program – **Google Chrome**
- Taken from Web artifact location –
(/img_drugs.E01/Users/shawn/AppData/Local/Google/Chrome/User Data/Default/History)

Artifact#41 – Transaction Related Search History 1			
Domain	bitcoinprice.com	Date & Time	2021-04-08 09:09:23 AWST (Date Accessed)
Title	Bitcoin Price (Today) – Live Bitcoin Price Charts & No Ads		
Link	https://www[.]bitcoinprice.com/		
Analysis	Users have searched for the price of bitcoin which could say that user intentionally searched to buy and sell illegally by using crypto currency.		
Artifact#42 – Transaction Related Search History 2			
Domain	zdnet.com	Date & Time	2021-04-08 08:30:45 AWST (Date Accessed)
Title	Coinbase: Possible ‘Significant’ drop in crypto prices this year like in 2018 ZDNet		
Link	https://www[.]zdnet.com/article/coinbase-significant-crypto-decline-similar-to-2018-is-a-possibility-this-year/		
Analysis	User has searched the rate of crypto currency which could say that user wanted to trade drugs online.		
Artifact#43 – Transaction Related Search History 3			
Domain	thevespiary.org	Date & Time	2021-04-08 09:08:25 AWST (Date Accessed)
Title	Cryptocurrency Price List – Top 20 Cryptocurrency Prices Today		
Link	https://www[.]bitcoinprice.com/list/		
Analysis	The user has visited a site about crypto currencies, which shows more involvement in cryptocurrencies.		
Artifact#44 – Transaction Related Search History 4			
Domain	commsec.com.au	Date & Time	2021-04-09 09:53:15 AWST (Date Accessed)
Title	What impacts share prices?		
Link	https://www[.]commsec.com.au/education/learn/choosing-investments/what-impacts-share-prices.html		
Analysis	The user has visited a site to read about share prices, which, combined with other evidence, suggests that the user might be researching financial markets, potentially for illegal financial transactions.		

11.0 ISSUE 4 - How Many Files Were Present in the System?

The rates below were taken from the overall investigation in the report.

File Type	Total Files on Device	Files Referenced in Investigation	Percentage (%)
Images	9,779	8	0.0818
Audio Files	303	0	0
Videos	387	0	0
Archives	57	0	0
Databases	93	0	0
Plain Text Files	396	7	1.77
Rich Text (RTF)	1,204	0	0
HTML Files	2,004	0	0
Office Files (Word, Excel, etc.)	45	0	0
PDF Documents	21	2	9.52
.exe Files	4414	6	0.14
.dll files	29029	0	0
.bat files	10	0	0
.com files	18	0	0
.cmd files	43	0	0
Deleted Files	7,480	0	0
Total	44,664	23	0.0515%

- ♦ A total of 44,664 files were present across various types, with 23 files referenced in the investigation. The percentage of files related to the offence is 0.0515%, indicating that only a small fraction of the total files is relevant to the illegal activity, but still the small proportion of files is very important. This was calculated without the online activity

12.0 ISSUE 5 - Was Any of the Installed or Previously Removed Software Used for the Alleged Crime?

All these files below were found from the path - /img_drugs.E01/Users/shawn/Downloads/, which is downloads folder from the Shawn's Directory.

Artifact Number& Name	Accessed Time	Software Download	App Name	The Software Purpose	Analysis
#45 Software 1	2021-05-13 11:13:02 AWST	PDFReaderSetup.exe	PDF Reader	To read and view PDF documents	The user installed a PDF reader, likely to read the pdfs in Artifact#9 and Artifact #10, which contained instructional information about marijuana cultivation.
#46 Software 2	2021-05-13 11:13:21 AWST	thebat_64_9-3-4.msi	The Bat (Email Client)	To manage email accounts securely and with encryption.	The user installed The Bat email client, which is typically very hard to trace. This suggests a possible interest in securely handling illicit communications.
#47 Software 3	2021-05-13 11:13:27 AWST	Thunderbird Setup 78.10.0.exe	Thunderbird (Email Client)	To manage and send encrypted email.	Thunderbird installation shows an interest in secure email management and encrypted communications. Potentially facilitate illegal exchanges.
#48 Software 4	2021-05-13 11:13:30 AWST	torbrowser-install-win64-10.0.12_en-US (1).exe	Tor Browser	To browse the internet anonymously and securely.	The user installed Tor Browser, could be to access the dark web as searched in Artifact#34.
#49 Software 5	2021-05-13 11:13:37 AWST	TrueCrypt-7.2 (1).exe	TrueCrypt	To encrypt and secure files and data on the system.	The installation of TrueCrypt indicates the user's interest in encrypting and protecting sensitive files, possibly securing sensitive data.
#50 Software 6	2021-05-13 11:13:38 AWST	VeraCrypt Setup 1.24-Update7.exe	VeraCrypt	To provide disk encryption for security and privacy.	VeraCrypt installation suggests an interest in securing files through encryption for data protection, possibly securing sensitive data.
#51 Software 7	2021-05-13 11:13:39 AWST	winrar-x64-600.exe	WinRAR	To compress, archive, and manage compressed files.	The user installed WinRAR, a tool for compressing and archiving files, possibly securing sensitive data.

13.0 Appendix A - Running Sheet

Date & Time (AWST)	Action Taken	Explanation	Outcome (Artifact No.)
Jan 6, 2024, 09:00:00	Launched Azure VM	The first step is to launch the virtual machine.	Virtual machine opened
Jan 6, 2024, 09:10:00	Opened Autopsy 4.21.0 and FTK Imager 4.5.0.3	I started forensic analysis using Autopsy 4.21.0 and FTK Imager 4.5.0.3, to examine digital evidence.	Evidence image loaded; initial file system structure visible.
Jan 6, 2024, 09:30:45	Used Data Sources >> drugs.E01_1 Host >> drug.E01 >> Shawn >> Downloads >> files in Autopsy 4.21.0	Examined files in the "Download" folder for potentially illegal content linked to drugs and firearms, including images, videos, and databases.	Artifact#1 to Artifact#17 found in the Download folder, potential links to illicit activities (excluding 8, 9, and 10).
Jan 6, 2024, 09:45:00	Using FTK Imager 4.5.0.3 , drugs.E01 >> New Volume [NTFS] >> Root folder >> Users Folder >> Shawn folder>> Downloads >> Files.	Cross-checked flagged files in FTK Imager to gather additional metadata and details, verifying their relevance to the case.	Artifact#1 to Artifact#17 cross-verified with FTK Imager, (excluding 8, 9, and 10).
Jan 9, 2024, 10:30:00	Used File Views → File Types → By Extension → Images, Videos, Audio, and Databases in Autopsy 4.21.0	Identified image files (Artifact#8) in Autopsy	Artifact#8 identified, flagged for further analysis.
Jan 9, 2024, 11:00:00	Used Autopsy 4.21.0 - Performed a Keyword Search for terms related to cryptocurrency (e.g., "Bitcoin," "wallet," "crypto").	Searched for cryptocurrency-related keywords to investigate potential illegal purchasing or money-laundering activities. Cryptocurrency is often linked to illicit transactions.	Artifact#20 - Email Web Artifact 2 found, detailing an email from Plus500 CFD: Online Trading to skemp19951@gmail.com.
Jan 9, 2024, 11:30:00	Used Autopsy 4.21.0, Performed a Keyword Search for terms related to “Google Account”	To check if there was a google account created with skemp19951@gmail.com, and to check when it was created.	Artifact#19 - Email Web Artifact 1 found, detailing the account creation email ("Shawn, finish setting up your new Google Account").
Jan 10, 2024, 10:30:00	Used Autopsy 4.21.0, Performed a Keyword Search for terms related to “Crypt”	To check if there are any crypto related transactions.	Found Artifact#18 as there are credit card numbers.
Jan 10, 2024, 11:30:00	Used File Views → File Types → By Extension → Executable → .dll files were searched in Autopsy 4.21.0 .	To recheck again and search for more.	Confirmed Artifact#18.
Jan 10, 2024, 12:30:00	Used Autopsy 4.21.0, Data Artifacts → Web Search	To check if the user searched related drugs.	Found Artifact#21 to Artifact#27

Jan 10, 2024, 15:30:00	Used Autopsy 4.21.0, Data Artifacts → Web History	Analyzed web history, cache, bookmarks, and cookies to find any illegal research on drugs or firearms, such as suspicious URLs, domain names, or search terms.	Artifact#29 to Artifact#40 identified through browsing history and flagged for further investigation.
Jan 15, 2024, 12:30:00	Used File Views → File Types → By Extension → Executable files → .exe in Autopsy 4.21.0	Examined executable to check for illicit software potentially linked to criminal activities.	Artifact#45 to Artifact#51 excluding (Artifact#46) identified and flagged based on functions of executable files.
Jan 15, 2024, 15:30:00	Used Data Sources >> drugs.E01_1 Host >> drug.E01 >> Shawn >> Downloads >> files in Autopsy 4.21.0	Checked the downloads folder to recheck and found the (thebat_64_9-3-4.msi)	The Aritfact#46 was found based on its function.
Jan 15, 2024, 12:30:00	Using FTK Imager 4.5.0.3 , drugs.E01 >> New Volume [NTFS] >> Root folder >> Users Folder >> Shawn >> Downloads folder >> Files	Cross checked all the files with FTK imager in the downloads folder.	Artifact#45 to Artifact#51 shows the installation of software related to this activity
Jan 16, 2024, 16:30:00	Used Autopsy 4.21.0 Performed a Keyword search for “Bitcoin”, “crypto”, “wallet” (Artifact#40 - Artifact#43)	Searched for terms related to cryptocurrency to investigate potential illegal purchases or transactions.	Found web artifacts related to Artifact#41 to Artifact#44
Jan 17, 2024, 14:30:00	Used Autopsy 4.21.0, Data Artifacts → Web History feature	Checked for titles found in web Artifact#41 to Artifact#44 in web history to find any crypto related searches.	Found web browsing history and related artifacts linked from Artifact#41 to Artifact#44
Jan 18, 2024, 10:00:00	Used Autopsy 4.21.0’s Keyword Search “LSD” and “MDMA” (Artifact#27)	Searched for drug-related terms like “LSD” and “MDMA” in the file system to find any evidence related to illicit drug activities.	Artifact#28 identified, possibly related to illegal drug activity.
Jan 21, 2024, 13:30:00	Used Autopsy’s File Views → File Types → By Extension → Documents → PDF	Investigated PDF files for any research or learning materials related to criminal activities like drugs and firearms.	Artifact#10 and Artifact #9 found based on relevant PDF file content.
Jan 24, 2024, 16:00:00	Final confirmation using FTK Imager 4.5.0.3 for all flagged files by checking the location of files in the pathway.	Used FTK Imager to verify the findings across all flagged files and ensure accuracy.	Artifact#1 to Artifact#51 rechecked for consistency and accuracy in findings.

14.0 Appendix B -Timeline Of Events

MARCH 2021: EXPLORATION OF DRUGS & MARIJUANA

March 22, 2021

09:16:46: [Artifact #21] - Drug Search: "Drugs"

- Analysis: Initial interest in drugs, indicating potential research..

09:18:21: [Artifact #22] - Drug Search: "Popular drugs among youth"

- Analysis: Exploring youth drug usage, suggesting a focus on trends for illicit activity.

09:19:23: [Artifact #23] - Drug Search: "Most common drugs among youth"

- Analysis: Further narrowing focus on commonly used substances among younger people.

09:20:21: [Artifact #24] - Drug Search: "Most common drugs among youth in Australia"

- Analysis: Could be planning to sell within the country.

09:26:12: [Artifact #25] - Drug Search: "Most expensive drugs"

- Analysis: Interest in high-value drugs, potentially related to trafficking or market research.

09:27:26: [Artifact #26] - Drug Search: "Marijuana"

- Analysis: User explicitly focused on marijuana, possibly for illegal cultivation.

March 25, 2021

09:17:44: [Artifact #27] - Drug Search: "Different types of drugs and their prices"

- Analysis: Deepening interest in the economic aspect of the drug market..

09:38:25: [Artifact #33] - Web Visit: ILoveGrowingMarijuana.com

- Analysis: User researching marijuana cultivation, likely for personal or illegal distribution.

09:39:33: [Artifact #32] - Web Visit: Marijuanaplantsonline.com

- Analysis: Visiting a site to purchase marijuana plants, indicating a potential move toward drug cultivation

09:41:38: [Artifact #19] - Account Creation: Email Title: "Shawn, finish setting up your new Google Account"

- Analysis: New account setup for potentially secure, anonymous communication related to illicit activities.

APRIL 2021: RESEARCH & TOOLS FOR ILLICIT ACTIVITIES

April 8, 2021

09:09:23: [Artifact #41] - Transaction-Related Search: BitcoinPrice.com

- Analysis: The interest in cryptocurrency for transactions, likely related to illicit drug trade.

09:57:02: [Artifact #27] - Drug Search: "How to grow LSD, MDMA, and LSD"

- Analysis: User researching how to manufacture illicit drugs.

April 9, 2021

11:39:12: [Artifact #31] - Web Visit: thevespiary.org (Cannabis Grow Bible)

- Analysis: Research into cannabis cultivation methods, likely for illegal purposes.

11:56:23 : [Artifact#39] - Searched for Bitlocker

- The search for BitLocker shows an effort to decrypt secure data, likely related to illegal or sensitive files

12:39:32: [Artifact #30] - Web Visit: Weedmaps.com

- Analysis: Website for locating marijuana dealers; the user may be attempting to contact dealers.

April 29,2021

11:01:32 AWST : [Artifact #38] - Searched for ritlabs

- Analysis: Searching for a secure email client, shows a continued effort for untraceable communication

MAY 2021: READING & ENCRYPTION TOOLS

May 13, 2021

11:12:51: [Artifact #10] - Reading PDF: "The Marijuana Grow Bible.pdf"

- Analysis: User is likely using the PDF for guidance on marijuana cultivation, supporting the hypothesis of an interest in illegal growing operations.

11:13:02: [Artifact #45] - Software Installation: PDFReaderSetup.exe

- Analysis: The installation of PDF reader software could indicate the user is preparing to access illicit manuals such as the Artifact#10

11:13:21: [Artifact #46] - Software Installation: The Bat (Email Client)

- Analysis: Installation of a secure email client as searched shows the user's need for encrypted communications.

11:13:37: [Artifact #49] -Software Installation: Veracrypt

- Analysis: VeraCrypt installation suggests an interest in securing files through encryption for data protection, possibly securing sensitive data.